Summary of Changes

Distance-Preserving Digests: A Primitive for BFT Consensus (BLOCKCHAIN'26 camera-ready)

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References give the location in the submitted IEEEtran version first and in this Springer camera-ready second; condensation to the 10-page limit turned several IEEE subsections into run-in paragraphs inside renumbered sections.

Reviewer 1

1. No heading directly followed by another heading. Every section now opens with a substantive lead-in paragraph stating what problem the section addresses and how its parts connect. Lead-ins were added at the former junctions: the primitive (was Sect. III, now Sect. 4), the consensus protocol (was Sect. IV, now Sect. 5), cross-shard consistency (was Sect. V, now Sect. 6), the evaluation (was Sect. VII, now Sect. 7), and the security analysis (was Sect. VIII, now Sect. 8).

2. Related work should follow the introduction. Related work moved from the back (was Sect. IX) to directly after the introduction (now Sect. 2). The redundancy this exposed with the introduction (LSH novelty, Ethereum committee mechanics, cross-shard coordination detail) was merged into the relocated section; the trims are logged in `CUTS.md` in the repository.

3. Add implementation details. An implementation paragraph now closes the evaluation (was Sect. X, now the "Implementation" paragraph of Sect. 7): digest computation cost and layout (one SHA-512 plus eight modular reductions per transaction, vectorized sums, 64 bytes on the wire), the two interchangeable BLS code paths (real BLS12-381 via py-ecc in the demo, hash-based mocks preserving the 96-byte signature and 48-byte public-key wire sizes in benchmarks) with their cross-validation (identical per-block message totals on the small-N demo under both), and the aggregation-side optimizations in the codebase (single-push bloom-diff sync, direct aggregation of speculative Phase 1 commitments into the fast-path certificate, per-leaf splitting of tree aggregation). Only what exists in the cited repository is described.

Reviewer 2

1. Abstract: the 8-dimensional digest space needs background. The abstract now explains the object before using it: each transaction hashes to a point in an 8-dimensional space, a validator's digest is the sum of its transactions' points, and the Euclidean distance between two digests is therefore proportional to the number of differing transactions. The abstract stays within 150–250 words.

2. Motivate the static adversary model. The introduction's closing paragraph now states that static corruption is the standard first-analysis setting for this protocol class, the model in which the classical safety arguments for PBFT, HotStuff, and Tendermint are stated, and that it isolates the primitive's contribution from adaptive corruption; adaptive adversaries are declared out of scope with a pointer to the limitations discussion (now Sect. 9).

3. Sect. III.D numbers must be derivable. The liveness bound (now the "Threshold calibration and liveness bound" paragraph of Sect. 4) derives every constant in prose: $k_{\max}$=2 is the partial-observation model the calibration samples; the sampled 99th percentile ≈4.1 times the 1.2 margin gives $\tau \approx 4.9$; the conditional exceedance 0.005 is measured over the same sample; multiplying by $p_{\text{miss}} = 0.37$ gives $p = 0.00185$; Hoeffding with $t = 1/3 - p \approx 0.33$ (so $2t^2 \approx 0.22$) yields $\exp(-0.22N)$, evaluated as $\exp(-22) \approx 3{\times}10^{-10} < 10^{-9}$ at N=100 and $\exp(-220) < 10^{-95}$ at N=1000.

4. Overview paragraph for the protocol section. Sect. 5 (was Sect. IV) opens by stating what each phase does, that the phases collapse into one round trip when enough validators already hold the block, and why the signature phase can be accelerated but never skipped.

5. Overview paragraph for the evaluation section. Sect. 7 (was Sect. VII) opens with the three questions the evaluation answers (message load at scale, the BLS bottleneck, the cost to Byzantine tolerance) and the measurement discipline (a counter on every simulated send, identical byte constants across protocols).

6. Informal phrasing in the cross-shard section. In Sect. 6 (was Sect. V): "Lower latency, but receipts scale with transaction volume" became "...achieves lower latency than two-phase commit at the price of receipt volume that grows linearly with cross-shard transaction volume"; "tells you identical or not-identical" became "hash-based verification indicates only whether two states are identical, revealing nothing about the magnitude of divergence." The section's remaining fragments (the table lead-in, the multi-shard enumeration) were rewritten as complete sentences.

7. HotStuff referenced without introduction. At first mention (Sect. 1) HotStuff is now described self-contained: a leader-based BFT protocol whose rotating leader drives each block through three voting rounds, collecting each round's votes into a threshold-signature quorum certificate proving $2N/3$ support. Later references build on this; nothing beyond what the comparisons use was added.

Additional changes

(a) Template. Reformatted from IEEEtran two-column to the Springer template required by the congress (`llncs` class, `splncs04` bibliography, keywords after the abstract, no page numbers), renumbering all sections, figures, and tables.

(b) Author-identified fast-path correction. The submitted version issued the fast-path certificate from Phase 1 digest agreement alone (near-zero cluster variance), backed by no signatures, inconsistent with the paper's own analysis (Theorem 1's premise is $2N/3$ signed commitments; Sect. VIII.D states Phase 1 contributes no cryptographic security). The mechanism is repaired, not softened: Phase 1 messages now carry a BLS commitment on $(\mathit{height}, H(B_{\text{prop}}))$, attached iff the validator's local state matches the proposal (64+25+96-byte payload from matching validators), and the fast path triggers only when the aggregator collects $\geq 2N/3$ valid commitments on one hash in round one, aggregating them into a certificate byte-identical to the Phase 2 object (96-byte aggregate plus $N/8$-byte bitmap). Single-round finality is thus covered by Theorem 1 verbatim, and an aggregator can at most withhold a certificate (liveness), as noted after Corollary 1. On a miss, the round falls back to full Phase 2 collection exactly as submitted; Phase 1 commitments may be reused there, a saving the evaluation conservatively does not model. The fast path applies to the flat protocol; carrying speculative commitments up the tree is deferred. $(1 - p_{\text{miss}})^N$ is now labelled a lower bound on engagement (matching all honest validators, where only $2N/3$ commitments are required). The mechanism follows the speculative lineage of Zyzzyva (Kotla et al., SOSP 2007) and SBFT (Gueta et al., DSN 2019), both now cited. The Byzantine-sweep figure (was Fig. 9, now Fig. 2) was regenerated by re-running the repository's figure script so its bandwidth panel reflects the 96-byte Phase 1 commitment; a pinned-clock comparison verified that success rates, message counts, and the other protocols' bandwidth are bit-identical, and at $p_{\text{miss}}=0.37$ the fast path does not engage, so all other reported numbers are unchanged. The repository implementation was updated to match, and the protocol-flow figure was redrawn (now script-generated in the repository) so its decision node, Phase 1 message contents, and fast-path terminal show the corrected mechanism.

(c) Condensation. The mechanical Springer conversion ran to 17 pages before the requested additions, so the camera-ready is condensed to the 10-page limit: secondary derivations became one-line results with pointers to the extended version (technical report at `https://github.com/RyanMercier/Proxima`), figures whose plotted values are fully stated in the retained tables and prose were removed, the messages and latency tables were merged, and several subsections became run-in paragraphs. No reviewed number, table value, or plotted datum was changed; no technical content beyond (b) was added. The full cut log is `CUTS.md`.